

Math 300

Quiz # 7

I will not cheat on this quiz. _____

1. (2 pts) Give examples of two **finite** sets with different cardinalities.

2. (2 pts) Give examples of two **infinite** sets with different cardinalities.

3. (4 pts) Consider the closed interval $[0, 5]$.

a) Give an example of a partition, P , of this interval into 5 subintervals if the width of your largest subinterval is 2.

$P =$

b) Give a refinement, Q , of your partition in part a.

$Q =$

4. (2 pts) Let $A = \left\{ 1 + \frac{1}{n} \mid n \in \mathbb{N} \right\}$. What are the supremum and the infimum of A ?

$\sup(A) =$ _____

$\inf(A) =$ _____

OVER $\rightarrow$

5. (2 pts) Give an example of an uncountable set that is bounded above but does not contain its supremum.

6. (2 pts) Identify the first few decimal digits of a real number b that could result from applying Cantor's second diagonalization method to the sequence of real numbers given below.

$$a_1 = 0.1010999.....$$

$$a_2 = 0.0123456....$$

$$a_3 = 0.3333333....$$

$$a_4 = 0.5656565 \dots$$

$$a_5 = 0.2500000....$$

$$a_6 = 0.1111111....$$

7 (2pts) The first few sets in a sequence defined by a one-to-one correspondence, g , mapping the natural numbers N to the power set $P(N)$ are shown in the list below. Identify the first few elements of the set D that Cantor would use to show that this function g cannot map onto all of $P(N)$.

$$g(1) = \{2\}$$

$$g(2) = \{1,2,4,8,16,32,.....\}$$

$$g(3) = \{3\}$$

$$D =$$

$$g(4) = \{1,2,3\}$$

$$g(5) = \{1,3,5,7\}$$

$$g(6) = \{\}$$

8. (4 pts) List four topics that you enjoyed learning about in this course.

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